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Research paper

Performance of a novel axisymmetric wave-concentrator-integrated oscillating water column system

Yuhao Cen [a,b], Qianze Zhuang [c], Dongfang Liang [a], Tahsin Tezdogan [b], Xiaodong Liu [d], Yifeng Yang [e,f,g,*]

[a] Department of Engineering, University of Cambridge, Cambridge, CB2 1PZ, UK
[b] Department of Civil, Maritime and Environmental Engineering, Faculty of Engineering and Physical Science, School of Engineering, University of Southampton, Southampton, SO16 7QF, UK
[c] School of Infrastructure Engineering, Dalian University of Technology, Dalian, 116024, China
[d] College of Ocean Science and Engineering, Shandong University of Science and Technology, Qingdao, 266590, China
[e] State Key Laboratory of Subtropical Building and Urban Science, South China University of Technology, Guangzhou, 510641, China
[f] School of Civil Engineering and Transportation, South China University of Technology, Guangzhou, 510641, China
[g] Department of Mechanical Engineering, University College London, London, UK

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ABSTRACT

Inspired by metamaterial concentrators for electromagnetic (EM) waves, water wave concentrators have recently been invented to achieve localised wave height amplification, offering a promising pathway for enhancing wave energy conversion efficiency. This study proposes the integration of an oscillating water column (OWC) wave energy converter (WEC) with an annular wave concentrator, achieving significant performance enhancement through an axisymmetric omni-directional design. A computational fluid dynamics (CFD) model is developed to evaluate the energy capture performance of the integrated system after being validated against experimental data. The effects of wave nonlinearity and baffle configuration on the hydrodynamic efficiency are subsequently investigated. The results demonstrate that integrating the wave concentrator significantly enhances the hydrodynamic efficiency, yielding a 77.3% increase in the maximum capture width ratio (CWR) compared with a standalone OWC. Notably, the efficiency is enhanced several times for long-period waves (2.8–3.8 s), owing to wave refraction and superposition within the concentrator that amplify the free surface oscillation inside the chamber to nearly twice the incident wave amplitude. Wave nonlinearity further enhances the energy conversion rate near the resonant frequency, yielding an additional 43.5% efficiency improvement as the wave height increases from 0.050 m to 0.100 m. Increasing the number of guiding baffles can enhance the hydrodynamic efficiency through effective flow channelling, although the efficiency gain gradually levels off at high baffle densities.

1. Introduction

In recent decades, escalating energy demand and persistent shortages in energy supply have become pressing global concerns. Electricity generation systems are increasingly strained, underscoring the urgent need for clean, secure, and resilient power sources to support sustainable development (Adcock et al., 2021; Capellán-Pérez et al., 2014). Among the available renewable energy options, ocean wave energy has attracted growing attention owing to its high energy density, reliability, and strong predictability (Li et al., 2025; López et al., 2013). Despite this promise, wave energy conversion technologies remain at an early stage of development, with only a limited number of systems achieving commercial deployment (Boodoo et al., 2026; Tiron et al., 2015). This gap highlights the necessity for continued research, optimisation, and technological advancement.

A wide variety of wave energy converters (WECs) have been proposed and tested, employing different power take-off (PTO) mechanisms to transform wave-induced motion into useable electrical energy (Ahamed et al., 2020; Paduano, 2026). Among these concepts, the oscillating water column (OWC) stands out for its structural simplicity,

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* Corresponding author.
E-mail address: yifeng.yang.19@ucl.ac.uk (Y. Yang).

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robustness, and relatively low operational and maintenance requirements (Alam Towhid et al., 2026; Machado et al., 2026; Romolo et al., 2024). An OWC typically consists of a partially submerged hollow chamber that traps a column of water (Falcão and Henriques, 2016). As incident waves cause the water column to oscillate vertically, the resulting bidirectional airflow is forced through an opening at the top of the chamber (Batlle Martin et al., 2023). This airflow drives an aerodynamically coupled turbine-generator system, enabling the conversion of wave energy into electrical power. In recent years, the OWC technology has progressed from conceptual development toward practical deployment, as demonstrated by a limited number of full-scale implementations (Carrelhas and Gato, 2023). A prominent example is the Mutriku Wave Power Plant, the world's first multi-turbine OWC facility integrated into a breakwater. Since its installation, the plant has been operating continuously, and by 2020 it had achieved a cumulative electricity production of 2 GWh, representing a significant milestone for commercial wave energy applications (Renzi et al., 2021). In parallel, advanced OWC concepts have also been tested at full scale. Based on U-OWC technology, the Resonant Wave Energy Converter 3 (REWEC3) incorporates a vertical U-shaped duct designed to enhance hydraulic coupling between the oscillating water column and the open wave field, and it has been deployed as a full-scale prototype in the Port of Civitavecchia (Arena et al., 2013). Despite these technological advancements, the widespread deployment of wave energy systems remains constrained. A primary challenge lies in enhancing the overall energy conversion efficiency of OWCs, which is essential for improving their commercial viability (De Andres et al., 2017).

In response to this challenge, substantial research efforts have been devoted to improving OWC performance. Rezanejad et al. (2019) showed that the incorporation of a stepped seabed configuration can lead to an improvement in the performance of OWC-based WECs. Abbasi and Ketabdari (2022) reported a 2.3% improvement in efficiency by applying micrometre-scale riblet structures to the blades of a Wells turbine in an OWC system. In addition to geometric optimisations, operational strategies have also been shown to influence OWC performance. Liu et al. (2022) found that employing a free-spinning turbine mode results in higher efficiency for an OWC system. The use of multi-chamber configurations has been identified as an effective approach for improving the capture width ratio (CWR) of OWCs. Taherian Haghighi et al. (2021) conducted a comparative study between single- and dual-chamber OWCs and demonstrated that the dual-chamber configuration not only increases the peak hydrodynamic efficiency but also broadens the effective operating bandwidth, particularly at shorter wavelengths. Wang et al. (2021) developed a theoretical model for a dual-chamber OWC device incorporating a pitching front lip-wall and demonstrated that an asymmetric structural configuration can lead to improved hydrodynamic efficiency. Despite the demonstrated benefits of geometric and operational optimisations, enhancing OWC performance solely through modifications of the standalone device remains challenging, motivating the exploration of alternative and integrated design approaches (Wang et al., 2023).

Beyond standalone OWC configurations, integrating OWCs with other structures, such as breakwaters and porous elements, has also been shown to enhance their hydrodynamic performance. Zhuang et al. (2024) conducted experimental and numerical investigations on a land-fixed OWC integrated with a porous plate, reporting improved energy conversion efficiency under low-frequency wave conditions. This enhancement was attributed to the porous plate promoting the transmission of first-order wave energy into the chamber while suppressing the transfer of wave energy to higher-order wave components. Zhang et al. (2025) proposed a cylindrical dual-chamber OWC-WEC combined with a parabolic breakwater and observed significantly increased air pressure together with approximately threefold amplification of wave elevation within the chamber. Li et al. (2026) demonstrated that integrating a local arc-shaped reflector and a horizontal bottom plate into a cylindrical OWC can substantially enhance its hydrodynamic efficiency, achieving a maximum increase of approximately 130% compared with the original configuration. Cheng et al. (2022) developed a hybrid multi-purpose concept by deploying an oscillating buoy within the chamber of an OWC and integrating the combined system into a π-type floating breakwater, demonstrating enhanced wave energy conversion performance together with improved wave attenuation. While these integration strategies have shown promising performance improvements, their effectiveness is generally limited to unidirectional wave incidence. In realistic ocean environments, wave fields are inherently multi-directional and irregular, which significantly limits the practical applicability of such direction-sensitive designs (Liu et al., 2023).

Recently, metamaterial-inspired structures have attracted increasing attention for their ability to manipulate and focus wave energy within designated regions, providing new opportunities to enhance wave energy conversion efficiency (Zhu et al., 2024). Metamaterials, initially introduced for electromagnetic applications, have been widely applied across multiple wave systems, such as acoustic, elastic, and water waves (Chen and Chan, 2007; Pendry et al., 2006; Schurig et al., 2006; Stenger et al., 2012). Within the field of water wave manipulation, a wide variety of functional wave-control devices have been proposed, enabling capabilities such as wave cloaking (Zareei and Alam, 2015), field rotation (Chen et al., 2009), energy concentration (Li et al., 2018), attenuation (De Vita et al., 2021), wavelength modulation (Zhang et al., 2023), and wave polaritons (Han et al., 2022). Among these, wave concentrators (WCs) constitute a distinctive category of metamaterial-inspired structures aimed at achieving localised wave height amplification (Saifullah et al., 2025). Cen et al. (2024a) provided a theoretical explanation for the transformation of wave patterns around a shallow-water waveguide by making an analogy between the shallow-water equations and Maxwell's equations, demonstrating that such a structure can significantly amplify wave heights above the waveguide platform. Building on this concept, an annular wave concentrator was subsequently investigated, in which local variations in water depth induced wave refraction and superposition, thereby enabling direction-independent wave energy focusing (Cen et al., 2024b). Inspired by metamaterial superscatterer concepts, a multi-layered cylindrical structure has recently been reported to exhibit both wave focusing and downstream sheltering effects in regular waves, suggesting its potential for integrated breakwater–WEC system design (Cen et al., 2025). In addition to metamaterial-inspired concepts, wave manipulation through periodic seabed topography has also been extensively investigated. Notably, recent studies on Bragg resonant reflection have provided valuable insights into how periodically varying bathymetry can effectively reflect and redistribute incident wave energy, thereby mitigating resonance phenomena in coastal environments (Gao et al., 2024a; Hou and Gao, 2026).

Despite the demonstrated wave focusing capability of the existing metamaterial-inspired structures, their systematic integration with wave energy converters and the resulting implications for energy conversion performance remain largely unexplored. In particular, previous studies have primarily focused on wave field manipulation itself, while the coupling mechanisms between metamaterial-inspired wave concentration and the internal dynamics of WECs have not been adequately addressed (Saifullah et al., 2025; Zhu et al., 2024). The specific contributions of this study are summarised as follows.

(1) A novel axisymmetric wave energy conversion concept is proposed by integrating a cylindrical OWC with an annular wave concentrator, enabling enhanced, wave-direction-independent energy capture, which is inherently desirable for deployment in realistic multi-directional wave environments.
(2) The coupling mechanisms between metamaterial-inspired wave concentration and OWC internal hydrodynamics are systematically revealed, demonstrating how wave–wave and wave–structure interactions intensify free surface oscillations and

chamber air pressure, thereby enhancing energy conversion efficiency.

(3) A systematic parametric investigation is conducted to quantify the influence of wave conditions and structural configurations on the hydrodynamic efficiency of the integrated system, providing practical guidance for performance-oriented design and optimisation.

The remainder of this paper is organised as follows. Section 2 introduces the design of the integrated WC–OWC system, the numerical model, and the data analysis method. Section 3 first presents the results of the mesh independence study and the validation of the numerical model against experimental data. This is followed by a comparative analysis between the single OWC and the integrated WC–OWC systems. The remaining part of this section is devoted to investigating the effects of wave nonlinearity and structural design parameters on the wave energy conversion performance. Finally, the main conclusions of this study are summarised in Section 4.

2. Methodology

2.1. Integrated system design

Inspired by metamaterial concentrators for electromagnetic (EM) waves, an annular water wave concentrator has been proposed to amplify wave heights within a targeted region (Cen et al., 2024b; Li et al., 2018). By locally modifying the water depth and flow direction, the concentrator focuses wave energy into a small area. Given the inherently dispersed nature of ocean wave energy distribution, placing WECs in regions where wave energy is concentrated by the WC is expected to enable more efficient and cost-effective energy harvesting.

In this study, the annular WC is integrated with a cylindrical OWC device to investigate the wave energy capture performance of the combined configuration. A schematic diagram of the integrated WC–OWC system is shown in Fig. 1. A Cartesian coordinate system (x, y, z) is adopted, where the positive x-direction coincides with the incident wave propagation direction, the y-direction is lateral, and the positive z-direction is vertically upward. The origin of the coordinate system is located at the centre of the annulus at the still water level.

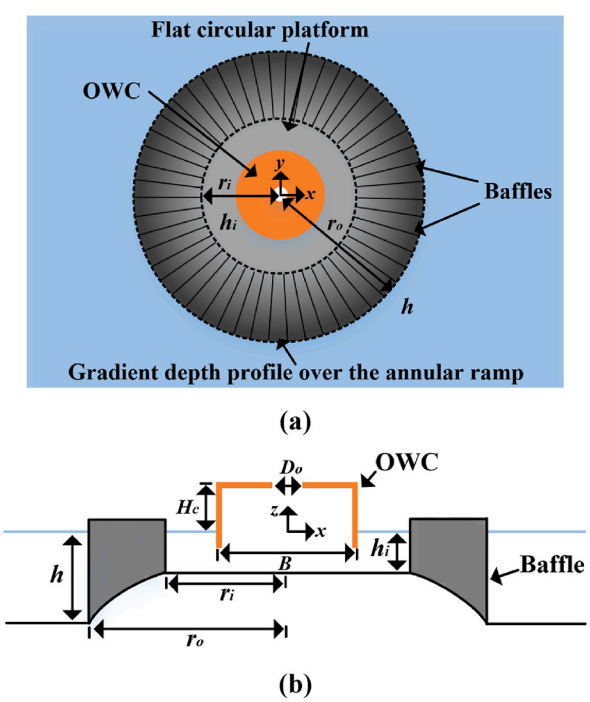

Fig. 1. Schematic diagram of the integrated WC–OWC system: (a) top view and (b) cross-sectional view.

The wave concentrator consists of three main components: fifty thin vertical baffles, an annular ramp, and a flat circular platform. The baffles have a uniform thickness of 0.01 m, are evenly spaced around the annulus, and extend above the still water level, thereby constraining wave propagation primarily to the radial direction. A gradient water-depth profile is implemented over the annular ramp, with the water depth distribution defined as (Cen et al., 2024b):

$$h_i = h(r_i/r_o)^2, r \le r_i, \tag{1}$$

$$h(r) = \frac{hr_i^2}{(r_o - r + r_i)^2}, r_i < r \le r_o, \tag{2}$$

where r_i, r_o are the inner and outer radii of the annulus, respectively, and h_i, h are the water depths above the central circular platform ($r \le r_i$) and outside the annulus ($r \ge r_o$), respectively. The water depth over the annular ramp ($r_i < r < r_o$) varies continuously from h_i to h in the radial direction.

The OWC model consists of a cylindrical chamber located at the centre of the wave concentrator. The chamber has an outer diameter B and a wall thickness b. A circular orifice is located at the roof of the chamber, with its centre coincident with the axisymmetric centre of the system. The opening ratio, defined as the ratio of the orifice area to the internal cross-sectional area of the air chamber, is set to 1.5%. The chamber draught d_c and air chamber height H_c define the submerged and air-filled portions of the chamber, respectively. The key geometric parameters of the integrated WC–OWC system are summarised in Table 1.

2.2. Numerical model

2.2.1. Governing equations

In this study, two key physical processes are considered: the gas–liquid coupling within the fixed cylindrical OWC during wave energy extraction, and the interaction between incident waves and the wave concentrator. CFD methods have been widely and successfully employed to investigate complex wave–structure interactions characterised by strong nonlinearity and resonance effects, such as transient gap resonance under focused wave groups (Gao et al., 2024b, 2026). The CFD model solves the three-dimensional incompressible Navier–Stokes equations to simulate both the air flow and wave motion, with air compressibility effects assumed negligible in the small-scale air chamber (Simonetti et al., 2018):

$$\frac{\partial u_i}{\partial x_i} = 0, \tag{3}$$

$$\frac{\partial u_i}{\partial t} + u_j\frac{\partial u_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left[\frac{\mu}{\rho}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)\right] + g_i, \tag{4}$$

where t is time, μ is the dynamic viscosity coefficient of the fluid, $x_i = (x_1, x_2, x_3)$ and $u_i = (u_1, u_2, u_3)$ are the Cartesian coordinates and flow

Table 1
Key parameters of the integrated WC–OWC system.

Component	Parameter	Value (m)
WC	Inner radius r_i	1.50
	Outer radius r_o	2.32
	Water depth above the platform h_i	0.25
	Water depth outside the annulus h	0.60
OWC	Chamber outer diameter B	1.50
	Chamber wall thickness b	0.01
	Chamber draught d_c	0.10
	Air chamber height H_c	0.40
	Orifice diameter D_o	0.18

velocities along the x, y and z directions, respectively, ρ is the fluid density, p is the pressure, and g_i is the acceleration due to gravity.

The interface between the air and water phases, i.e. the free surface, is captured using the volume of fluid (VOF) method. The two phases are assumed to be immiscible. In this approach, a volume fraction function $\alpha(x_i, t)$ is introduced for each computational cell in the domain. Based on this volume fraction, the fluid properties within each cell are defined as:

$$u_i = \alpha u_{iw} + (1 - \alpha) u_{ia}, \tag{5}$$

$$\rho = \alpha \rho_w + (1 - \alpha) \rho_a, \tag{6}$$

$$\mu = \alpha \mu_w + (1 - \alpha) \mu_a, \tag{7}$$

where the subscripts w and a represent the water and air phases, respectively. The volume fraction α is governed by the following transport equation:

$$\frac{\partial \alpha}{\partial t} + u_i \frac{\partial \alpha}{\partial x_i} = 0. \tag{8}$$

The volume fraction α equals 1 in cells fully occupied by water, equals 0 in cells filled with air, and takes intermediate values $(0 < \alpha < 1)$ in cells intersected by the gas–liquid interface. The free surface is represented by the iso-surface $\alpha = 0.5$.

2.2.2. Numerical wave flume

A three-dimensional numerical wave flume, as illustrated in Fig. 2, is established to simulate the hydrodynamic and aerodynamic responses of the integrated WC–OWC system. The WC–OWC model is positioned at a distance of 4λ (where λ denotes the wavelength) from both the inlet and outlet boundaries. Consequently, the total length of the wave flume varies with the wave period considered in this study. The flume height and width are set to 1.2 m and 7.5 m, respectively, with the latter corresponding to five times the OWC width. With this configuration, the influence of wave reflections from the flume sidewalls is considered to be minimal within the time frame of interest (Wang and Zhang, 2022).

Surface waves are generated at the inlet by applying a velocity inlet boundary condition. At each time step, the inlet velocity field and the corresponding free surface elevation are prescribed according to the fifth-order Stokes wave theory (Fenton, 1985). The incident wave conditions considered in the present study are summarised in Table 2, including the wave height H, wave period T, wavelength λ, and the dimensionless wave number kh. The top boundary of the flume is specified as an open-air boundary with atmospheric pressure. Only half of the computational domain is simulated by imposing a symmetry boundary condition at the geometric symmetry plane, which has been shown to significantly reduce computational expense without compromising solution accuracy (Li et al., 2026). The sidewalls and bottom of the flume, as well as the surface of the wave concentrator, are defined as slip walls, whereas a no-slip wall condition is applied to the surface of the OWC.

To minimise wave reflections and ensure the accuracy of the generated waves, a wave forcing zone is established near the inlet boundary. Within this zone, a forcing source term is added to the momentum equations to gradually relax the numerical solution toward the prescribed fifth-order Stokes wave solution:

$$q_\phi = -\gamma \rho (\phi - \phi^*), \tag{9}$$

where γ is the forcing coefficient that varies smoothly from zero at the end coordinate of the forcing zone in the flow direction to a recommended maximum value 10 s^{-1} at the inlet (Kim et al., 2012), ϕ is the current solution of the transport equation, ϕ^* is the value towards which the solution is forced.

To supress wave reflections at the outlet, a wave damping zone is introduced by adding a resistance term to the vertical momentum equation (Choi and Yoon, 2009):

$$S_z^d = -\rho(10 + 10|u_3|) \frac{e^\kappa - 1}{e^1 - 1} \omega, \tag{10}$$

with

$$\kappa = \left(\frac{x - x_{sd}}{x_{ed} - x_{sd}} \right)^2, \tag{11}$$

where x_{sd} and x_{ed} are the starting and end coordinates of the wave damping zone in the flow direction, u_3 is the vertical velocity of water.

Six wave gauges (G1–G6) are arranged within the OWC chamber to measure the free surface elevations. Gauges G4, G5, and G6 are located at a fixed distance of 0.2 m from the chamber centre, while the spacing between G1 and G4, G2 and G5, and G3 and G6 is set to 0.3 m. Air pressure inside the chamber is monitored using pressure sensors P1, P2 and P3 installed on the chamber roof, each located 0.6 m from the chamber centre. The arrangement of the wave gauges and pressure sensors is shown in Fig. 3.

The governing equations are discretised using the commercial CFD code Simcenter STAR-CCM+ (version 2406) based on the finite volume method (FVM). Temporal discretisation is performed using a second-order implicit backward Euler scheme. A time step of $T/800$ is adopted for the numerical simulations to ensure that the maximum courant number in the flow field of interest generally remains below 1. The velocity-pressure coupling is solved using the SIMPLE (Semi-Implicit Method for Pressure-Linked Equations) algorithm. The convection and diffusive terms are discretised with a second-order upwind scheme and a second-order central differencing scheme, respectively. To maintain a sharp interface between the two phases, the convective flux in the VOF transport equation is discretised using the High-Resolution Interface Capturing (HRIC) scheme. Turbulence effects are modelled using the Shear Stress Transport (SST) turbulence model, which combines the advantages of the k-ε model in the free stream and the k-ω model in the near-wall region.

2.3. Hydrodynamic efficiency

The conversion of wave energy into electrical energy proceeds through multiple successive energy conversion stages. The present study focuses only on the first stage, in which the incident wave energy is transformed into the kinetic energy of the oscillating air flow through the orifice of the OWC device. The hydrodynamic efficiency associated with this stage is typically characterised by the capture width ratio (CWR), defined as the ratio between the pneumatic power extracted by the OWC and the time-averaged energy flux of the incident waves.

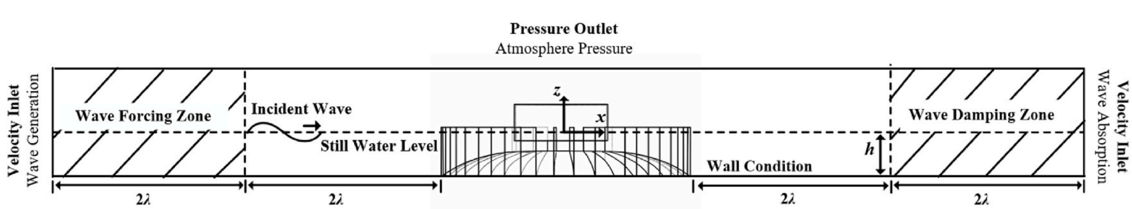

Fig. 2. Numerical wave flume setup.

Table 2
Incident fifth-order wave conditions considered in this study.

H (m)	0.050, 0.075, 0.100													
T (s)	1.2	1.4	1.6	1.8	2.0	2.2	2.4	2.6	2.8	3.0	3.2	3.4	3.6	3.8
λ (m)	2.13	2.71	3.28	3.83	4.37	4.91	5.43	5.95	6.47	6.98	7.49	8.00	8.51	9.02
kh	1.77	1.39	1.15	0.98	0.86	0.77	0.69	0.63	0.58	0.54	0.50	0.47	0.44	0.42

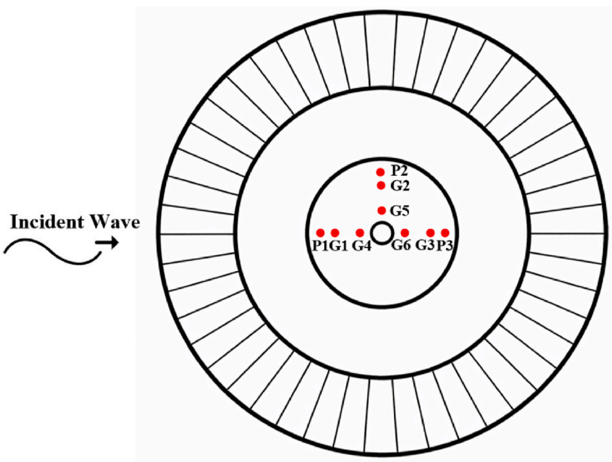

Fig. 3. Arrangement of the wave gauges and pressure sensors inside the OWC chamber.

Accordingly, the CWR is expressed as

$$\xi = \frac{P_{owc}}{P_w w}, \tag{12}$$

where P_{owc} is the pneumatic power absorbed by the OWC device, P_w is the average incident wave energy flux per unit crest width and w is the effective width of the incident waves interacting with the device. In the present configuration, w is taken as the internal diameter of the chamber, i.e. $B - 2b$, following common practice in studies of integrated OWC systems (Zhang et al., 2025; Zhuang et al., 2024). Although the wave concentrator modifies the incident wave field, it does not come into direct contact with the OWC chamber or directly participate in the air–water coupling process. Therefore, the energy conversion occurs within the chamber, and the chamber width is adopted as the characteristic length scale for the CWR calculation.

Assuming incompressible air flow, the pneumatic power extracted by the OWC model can be evaluated from the time-averaged product of the air volumetric flow rate through the orifice and the chamber air pressure:

$$P_{owc} = \frac{1}{T}\int_{t}^{t+T} Q(t)P_a(t)dt = \frac{\pi(B-2b)^2}{4T}\int_{t}^{t+T} v_c(t)P_a(t)dt, \tag{13}$$

where T is the incident wave period, Q is the air volume flux through the orifice per unit time, P_a is the air pressure inside the chamber, and v_c is the vertical velocity of the periodically oscillating water surface within the chamber. In the numerical simulations, v_c is obtained by differentiating the chamber free surface elevation, which is calculated as the average of the wave gauge measurements G1–G6, while P_a is determined by averaging the pressure signals recorded by sensors P1–P3.

It should be noted that the air inside the chamber is assumed to be incompressible in the present study. This assumption is considered reasonable for the current model-scale simulations, where the air pressure variation is relatively small. Previous studies have shown that the influence of air compressibility is negligible under such conditions (Li et al., 2026; Zhang et al., 2025). However, it is acknowledged that air compressibility may become significant under larger wave amplitudes or at prototype scale, where stronger air compression and expansion occur. Simonetti et al. (2018) demonstrated that neglecting air compressibility can lead to an overestimation of chamber pressure and capture width ratio, with the deviation increasing as the pressure amplitude rises. Similarly, Elhanafi et al. (2017) reported that the peak hydrodynamic efficiency at full scale may be up to 12% lower than that predicted under incompressible assumptions at model scale. In the present study, although different wave heights are considered, the selected wave conditions remain within a moderate range, for which the incompressible assumption is acceptable. Future work will consider incorporating air compressibility effects to further improve the accuracy of the predictions under larger wave amplitudes.

Based on the linear potential flow theory, the time-averaged energy flux per unit crest width of the incident wave is given by:

$$P_w = \frac{1}{8}\rho g H^2 C_g, \tag{14}$$

where H is the incident wave height, C_g is the incident wave group velocity defined as:

$$C_g = \frac{\omega}{2k}\left(1 + \frac{2kh}{\sinh 2\,kh}\right), \tag{15}$$

where h is the still water depth, ω is the angular frequency and k is the wave number. These parameters satisfy the dispersion relation:

$$\omega = \sqrt{gk \tanh kh}. \tag{16}$$

For unidirectional incident waves in open-sea conditions, the hydrodynamic efficiency of the OWC device, expressed in terms of the capture width ratio, is bounded within the range $0 \le \xi \le 1$. The lower bound corresponds to a condition in which no wave energy is extracted, whereas the upper bound represents an idealised lossless system.

3. Results and discussion

3.1. Model verification and validation

A mesh independence study is conducted for the WC–OWC model to determine an appropriate mesh resolution. Mesh generation is performed using the built-in automatic meshing capability of Simcenter STAR-CCM+, employing a trimmed cell mesher in combination with a surface remesher to generate predominantly unstructured hexahedral cells with trimmed elements adjacent to solid boundaries.

The influence of three mesh densities on the simulated free surface elevation and chamber air pressure is examined under identical wave conditions, with a fixed wave height of $H = 0.050$ m and a water depth of $h = 0.6$ m. The selected wave period $T = 3.2$ s corresponds to the resonant condition yielding the maximum CWR in the present study. Local mesh refinement is applied near the free surface and around the WC and OWC structures, with further refinement in the vicinity of the air orifice to accurately resolve small-scale air flow features. Near the free surface, the mesh resolution corresponds to approximately 80–120 cells per wavelength in the flow direction and 10–16 cells per wave height in the vertical direction. The cell sizes near the OWC chamber and the air orifice range from 0.0156 m to 0.005 m and from 0.0078 m to 0.0025 m, respectively. The details of the three mesh resolutions are listed in Table 3, with the total number of cells increasing from 11

Table 3
Details of the three mesh resolutions.

Mesh Resolution	Free surface along x axis	Free surface along z axis	OWC chamber (m)	Air orifice (m)	Number of cells (million)
Coarse	$80/\lambda$	$10/H$	0.0156	0.0078	11
Medium	$100/\lambda$	$14/H$	0.0078	0.0039	17
Fine	$120/\lambda$	$16/H$	0.005	0.0025	31

million to 17 million and 31 million. Fig. 4 shows the integrated WC–OWC model mesh for the medium-resolution case, which is presented as a representative example. Fig. 5 compares the time histories of the free surface elevation at G1 and the chamber air pressure at P1 obtained using different mesh densities. The free surface elevation η and air pressure P_a are normalised by A and ρgA, respectively, where ρgA represents the characteristic hydrostatic pressure induced by the incident wave of amplitude A. The results indicate that the medium-resolution mesh provides sufficient accuracy while maintaining reasonable computational cost and is therefore adopted for the following simulations in this study.

To date, no experimental study has been reported for the novel WC–OWC integrated system. However, the accuracy of the numerical model for a single water concentrator has been demonstrated in our previous work (Cen et al., 2024b). To further validate the present numerical framework, simulations are performed for a single cylindrical OWC model with identical geometry, except for a chamber draught of d_c = 0.2 m. The numerical predictions are then validated against the experimental data reported by Zhang et al. (2025).

The physical model experiments were conducted in a wave basin at the State Key Laboratory of Coastal and Offshore Engineering, Dalian University of Technology. The basin measures 40 m in length, 24 m in width, and has a maximum water depth of 1 m. A total of 70 piston-type wave paddles are installed at one end of the basin to generate waves, while a sloping beach is placed at the opposite end to absorb reflected waves. In addition, absorbing nets are arranged along both sidewalls to minimise reflection effects. The still water depth was set to $h = 0.6$ m. Regular waves with a constant height of $H = 0.050$ m and periods ranging from 1.2 to 2.6 s were considered in the experiments. The experimental setup is shown in Fig. 6, where the OWC is installed with its centre located 25.48 m downstream of the wave-maker paddles. For the single-chamber model, Orifice 2 was closed.

The time series of the predicted free surface elevations at gauges G1–G6 and the air pressure at sensor P1 are compared with the experimental measurements for a wave period of $T = 1.6$s, as presented in Figs. 7 and 8, respectively. Good agreement between the numerical and experimental results is observed. In addition, the numerical and experimental values of the capture width ratio ξ are compared with respect to the dimensionless wave number kh in Fig. 9, where $k = 2\pi/\lambda$ is the wave number, yielding a root mean square error (RMSE) of 0.016. Overall, the numerical model reproduces both the crest and trough values of the free surface elevation and air pressure with good accuracy. Moreover, it successfully captures the nonlinear characteristics of the air pressure variation, as evidenced by the pressure fluctuations near the zero-crossing points shown in Fig. 8. These results demonstrate that the present numerical model is capable of accurately resolving the nonlinear processes associated with wave energy extraction.

It should be noted that the present validation is conducted based on a standalone OWC model due to the lack of available experimental data for the integrated WC–OWC system. While this validation demonstrates the capability of the numerical model to accurately capture the key hydrodynamic and pneumatic processes within the OWC chamber, additional uncertainties may arise when extending the model to the integrated configuration. In particular, uncertainties may arise in (i) the prediction of wave height amplification within the concentrator, where factors such as grid resolution, interface-capturing accuracy of the VOF method, and numerical dissipation may influence the simulated free surface steepness and amplitude; (ii) the estimation of wave energy transmission into the chamber, which is sensitive to the resolution of the flow field near the chamber entrance and the accurate representation of wave reflection and diffraction around the outer wall. This becomes more complex in the presence of the wave concentrator, where the incident wave field is spatially redistributed due to wave focusing, affecting the effective energy flux entering the OWC; and (iii) the representation of air–water coupling within the chamber, where modelling assumptions such as incompressible air flow and the averaging of pressure signals from multiple measurement points may introduce deviations in the predicted pressure amplitude and phase relationship. These effects may become more pronounced under conditions with strong wave amplification induced by the wave concentrator. These factors may influence the accuracy of the predicted hydrodynamic efficiency, particularly in terms of peak magnitude. Nevertheless, the governing physical processes are consistently resolved within the same CFD framework, and the model is considered capable of providing reliable comparative assessments and quantitative analysis of the integrated system performance. Future work will focus on dedicated experimental validation of the WC–OWC configuration.

3.2. Performance comparison between the single OWC device and the WC–OWC integrated system

The capture width ratio serves as a primary metric for assessing the hydrodynamic efficiency of OWC systems. Accordingly, the CWRs of the single OWC device and the WC–OWC integrated system for $H = 0.050$ m

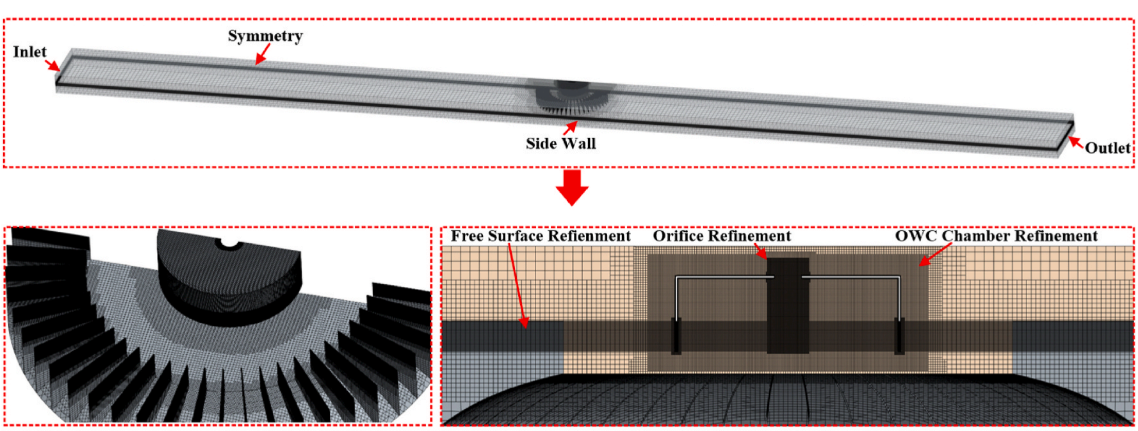

Fig. 4. Integrated WC–OWC model mesh for the medium-resolution case.

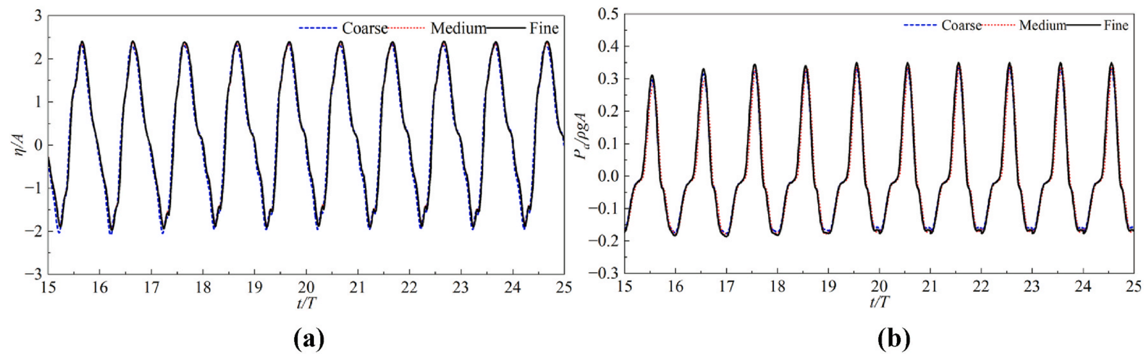

Fig. 5. Mesh independence test for the integrated WC–OWC model at $H = 0.050$ m and $T = 3.2$ s: (a) wave surface elevation at G1 and (b) air pressure at P1.

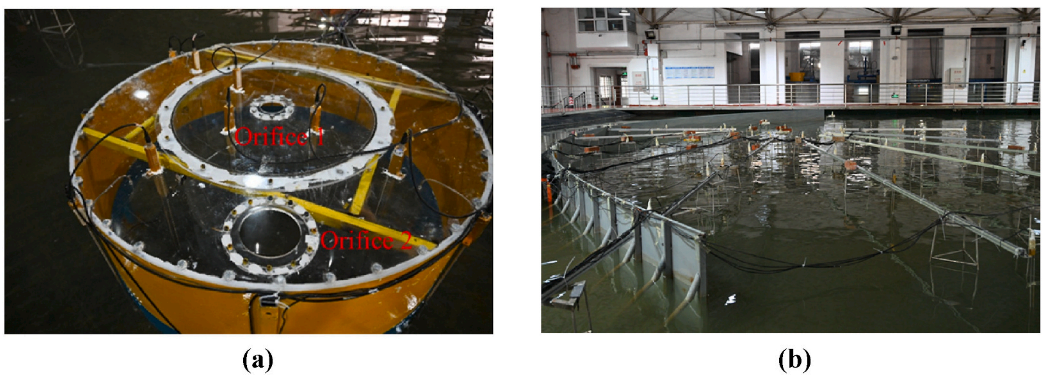

Fig. 6. Photographs of the experimental setup: (a) the OWC model and (b) the wave basin (Zhang et al., 2025).

are compared in Fig. 10 to highlight the performance advantages of the integrated configuration under an energy concentration environment. It should be noted that, in this comparison, the chamber draught of the OWC in both configurations is reduced to 0.100 m, rather than the 0.200 m used in the previous experimental setup. This adjustment is made to ensure sufficient clearance above the wave concentrator, thereby facilitating effective wave penetration into the OWC chamber. The analysis is conducted over a broad range of kh, varying from 0.42 to 1.77, which corresponds to wave periods from 1.2 s to 3.8 s with an increment of 0.2 s. The curves are fitted using the Akima spline interpolation, which is well suited for capturing complex trends in non-monotonic data without introducing spurious oscillations (Akima, 1970). This fitting approach is applied consistently in subsequent analyses.

For the single OWC device, the CWR exhibits a non-monotonic dependence on the dimensionless wave number kh. At relatively small values of kh, corresponding to longer waves, the OWC chamber behaves as a small perturbation to the incident wave field, allowing most of the waves to transmit past the structure while carrying the wave energy downstream. Consequently, only a limited amount of energy is captured within the chamber, resulting in a low CWR. As kh increases, wave penetration into the chamber is enhanced, leading to more effective excitation of the internal water column and a progressive increase in CWR until a peak value of $\xi = 0.24$ is reached at $kh = 1.39$. With a further increase in kh, corresponding to shorter waves, wave reflection at the external surface of the chamber wall becomes more pronounced, which reduces the amount of energy entering the chamber and causes the CWR to decrease. This behaviour agrees well with the experimental observations reported for a standalone single-chamber OWC device under similar wave conditions (Zhang et al., 2025).

In comparison, the WC–OWC integrated system exhibits a distinctly different variation in hydrodynamic efficiency. Within the low-kh wave regime ($kh = 0.42$–0.77), the presence of the wave concentrator significantly enhances wave focusing around the OWC chamber, resulting in a substantially higher CWR than that of the single OWC device. This demonstrates the ability of the integrated configuration to effectively capture energy from long waves that would otherwise bypass the standalone OWC. A pronounced peak efficiency of $\xi = 0.43$ is observed at $kh = 0.50$, representing an increase of 77.3% compared to the maximum CWR achieved by the single OWC device. As kh further increases, the wave focusing effect becomes less effective, and the interaction between the concentrated wave field and the OWC chamber weakens, leading to a reduction in energy capture efficiency. Nevertheless, a local peak at $kh = 1.15$ is observed and may result from a secondary resonance associated with the intrinsic resonant characteristics of the single OWC device. The underlying mechanism of this behaviour is further analysed in the following section based on the variations in the free surface elevation and the air pressure.

The observed shift of the resonant frequency toward lower values can be attributed to two primary factors. First, the wave concentrator itself exhibits a stronger wave amplification effect for long-period waves, as reported in previous studies (Cen et al., 2024b; Li et al., 2018). Second, the presence of the guiding baffles around the OWC chamber constrains the motion of the surrounding water, increasing the effective hydrodynamic added mass associated with the oscillating water column. This channelling effect further enhances its response to long waves and contributes to the low-frequency shift of the resonance response. A similar low-frequency shift of the resonant response has also been reported by Ning et al. (2019).

To further elucidate the mechanisms underlying the enhanced hydrodynamic efficiency of the integrated system, Fig. 11 presents the variations in the amplitudes of the free surface elevation and the air pressure inside the chamber. The free surface elevation amplitude, $\Delta\eta$, and the air pressure amplitude, ΔP_a, are calculated from the averaged wave gauge measurements (G1–G6) and the averaged pressure signals recorded by P1–P3 as $\Delta\eta = (\max(\eta) - \min(\eta))/2$ and $\Delta P_a = (\max(P_a) -$

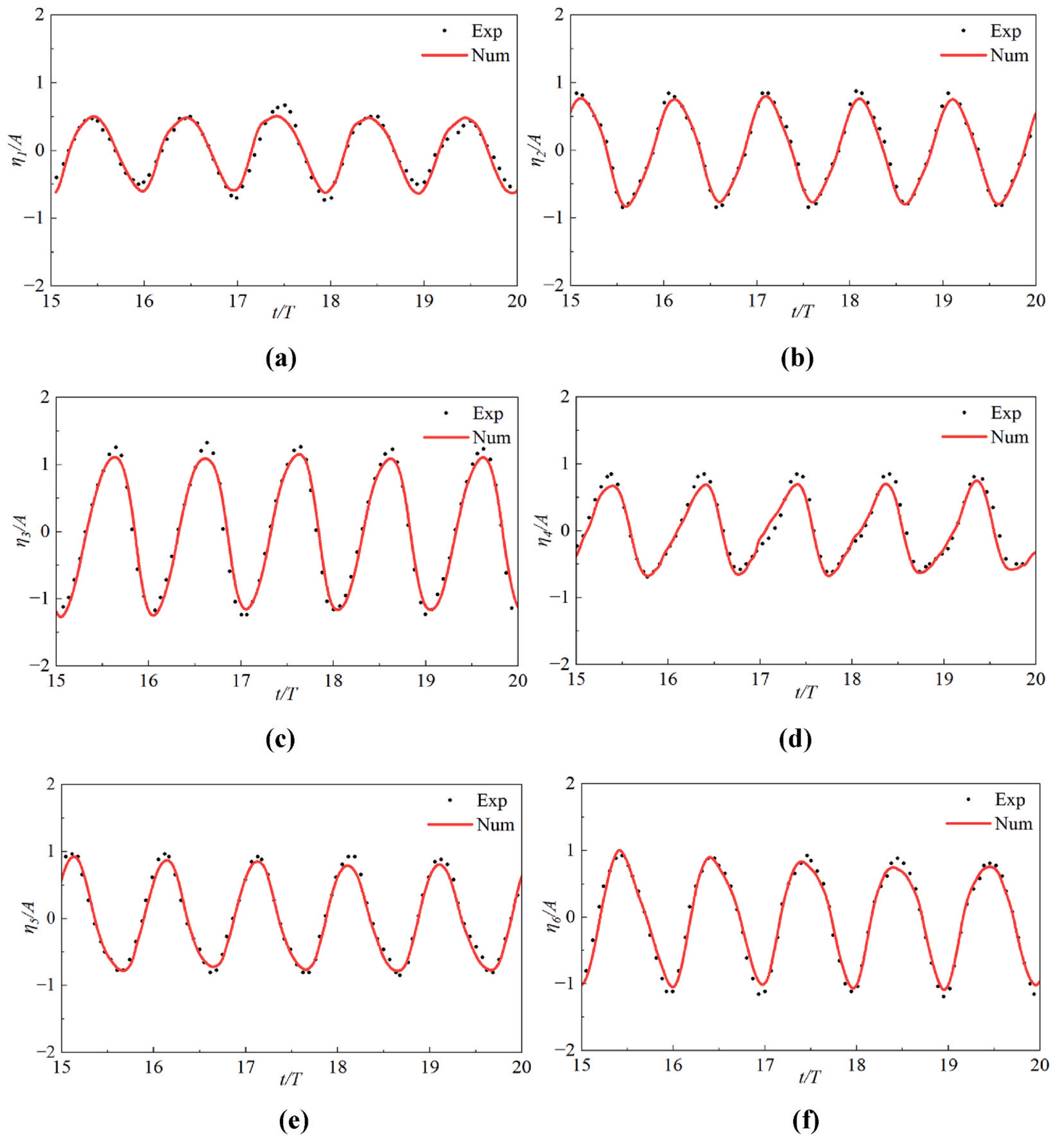

Fig. 7. Comparison of the free surface elevations between experimental and numerical results at (a) G1, (b) G2, (c) G3, (d) G4, (e) G5, and (f) G6 for $T = 1.6$ s.

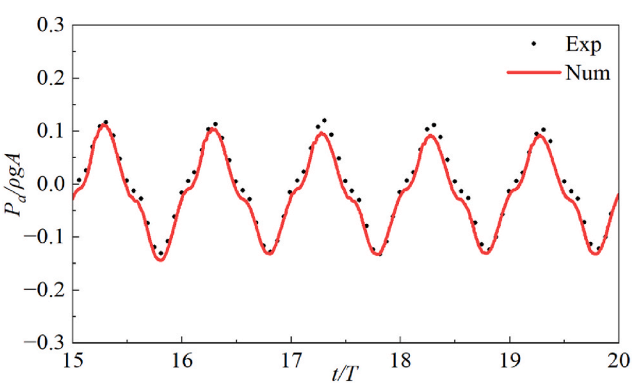

Fig. 8. Comparison of the air pressure between experimental and numerical results at P1 for $T = 1.6$ s.

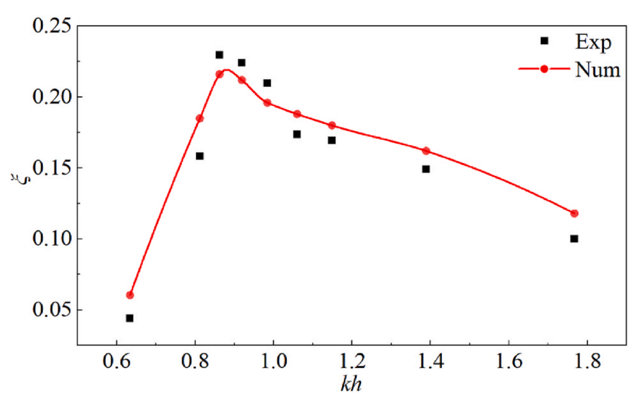

Fig. 9. Comparison of the hydrodynamic efficiency between experimental and numerical results for $T = 1.2$–2.6 s.

min $(P_a))/2$, respectively. It can be observed that, within the range of $kh = 0.42$–0.77, the integrated configuration exhibits pronounced wave amplification inside the chamber, with the free surface elevation reaching up to approximately twice the incident wave amplitude. In contrast, for the single OWC device, the free surface elevation amplitude within the chamber remains generally comparable to, or lower than, the incident wave amplitude over the same range of kh. This highlights the effectiveness of the wave concentrator in promoting wave focusing and

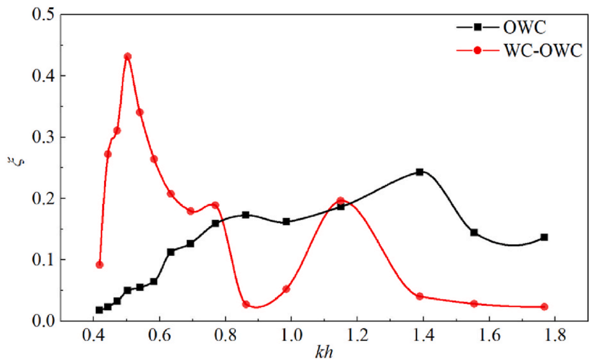

Fig. 10. Comparison of the hydrodynamic efficiency between the single OWC device and the WC–OWC integrated system as a function of *kh*.

penetration within the chamber under long wave conditions.

The amplified free surface elevation is accompanied by a corresponding increase in air pressure amplitude. A clear positive correlation is observed between the air pressure and free surface elevation responses, with both quantities exhibiting similar trends across the investigated range of *kh*. This phenomenon provides a direct explanation for the enhanced energy conversion performance of the integrated system observed in Fig. 10, particularly in the low-frequency wave regime. In addition, a local increase in both the free surface elevation amplitude and the air pressure amplitude is observed at $kh = 1.15$, compared to the neighbouring kh values. This behaviour suggests the occurrence of a local resonance-like response associated with the interaction between the wave concentrator and the intrinsic oscillation of the OWC chamber. It may be attributed to a higher-order hydrodynamic response induced by the wave concentrator, which enhances the dynamic coupling between the free surface motion and the chamber pressure variation under this condition. As a result, the energy transfer from the incident wave to the chamber is locally enhanced. However, the magnitude of this enhancement remains significantly lower than that associated with the primary resonance at $kh = 0.50$ (approximately 47% of the primary peak), indicating that its contribution to the overall energy conversion performance is comparatively less dominant.

Fig. 12 further quantifies the relative increase in hydrodynamic efficiency achieved by the WC–OWC integrated system. Although the integrated system exhibits limited or even negative gains at shorter wave periods, a substantial performance enhancement is achieved in the long-wave regime. Specifically, within the wave period range of 2.8–3.8 s, the relative increase in CWR reaches approximately 300%–1000%. This improvement arises primarily because the standalone OWC exhibits very low efficiency under long wave conditions, whereas the integrated configuration remains effective due to the wave energy concentration. These results highlight the advantage of the proposed system in altering the effective operating range of OWCs toward longer-period waves.

Fig. 13 shows the spatial distribution of the free surface elevation at the resonant condition of $kh = 0.50$, illustrating how incident waves are focused and effectively coupled with the OWC chamber. As the incident waves propagate over the wave concentrator, a clear transformation from straight to curved wavefronts can be observed due to wave refraction. This wavefront curvature leads to constructive interference among the refracted wave components, giving rise to a pronounced amplification of wave heights in the vicinity of the OWC chamber. Downstream of the integrated system, the wave gradually reverts back to its original plane wave pattern. Overall, the results demonstrate that the wave concentrator is able to enhance the hydrodynamic performance of the OWC system by promoting wave focusing and penetration at the chamber entrance.

3.3. Effects of wave height

To investigate the influence of wave nonlinearity on energy conversion performance, Fig. 14 compares the hydrodynamic efficiency of the WC–OWC integrated system under three wave height conditions, $H = 0.050$ m, $H = 0.075$ m and $H = 0.100$ m, over the dimensionless wave number range of $kh = 0.44 - 1.15$. It should be noted that, in addition to wave nonlinearity, variations in wave height also modify the relative relationship between the incident wave and the chamber geometry, particularly the draught-to-wave-height ratio, thereby influencing the wave transmission and energy capture process. The observed efficiency variation is therefore attributed to the combined effects of nonlinear wave behaviour and changes in wave–structure interaction associated with varying wave height. The present analysis only focuses on non-breaking wave scenarios, as numerical simulations indicate that pronounced wave breaking occurs inside the chamber when the wave height exceeds 0.100 m, and thus larger wave heights are not considered.

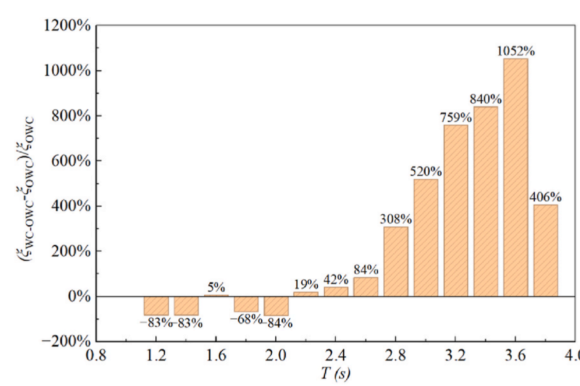

Fig. 12. Relative increase in hydrodynamic efficiency of the WC–OWC integrated system compared with the single OWC device for different wave periods.

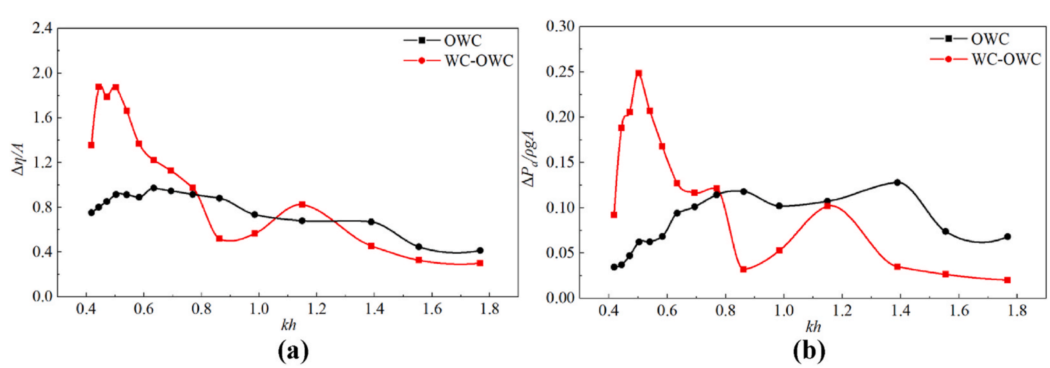

Fig. 11. Variations of the free surface elevation amplitude (a) and the air pressure amplitude (b) inside the chamber with *kh*.

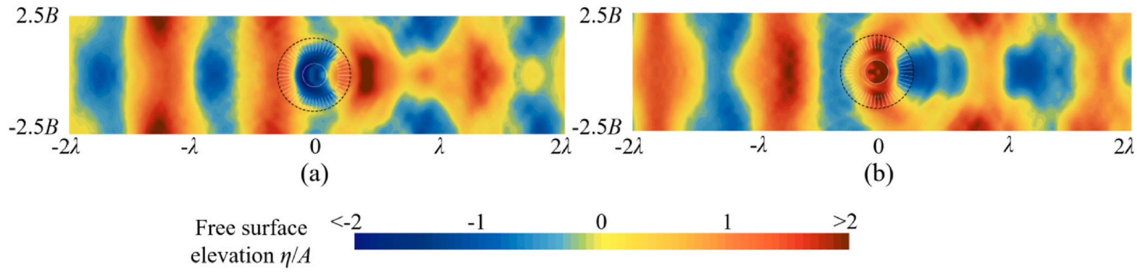

Fig. 13. Spatial distribution of the free surface elevation under the resonant condition $kh = 0.50$: (a) at $t = 20.0\ T$; (b) at $t = 20.5\ T$.

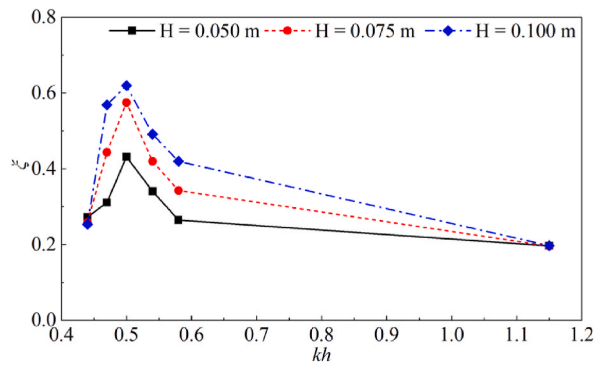

Fig. 14. Hydrodynamic efficiency of the WC–OWC integrated system for $H = 0.050$ m, $H = 0.075$ m and $H = 0.100$ m.

The results show that an increase in wave height does not alter the resonant condition of the integrated system, which remains at $kh = 0.50$. As the wave height increases, the CWR generally exhibits an increasing trend, except at $kh = 0.44$ and $kh = 1.15$. In particular, at $kh = 0.50$, the CWR increases by 33.1% and 43.5% for $H = 0.075$ m and $H = 0.100$ m, respectively, compared with the case of $H = 0.050$ m, reaching a maximum value of 0.62. This indicates that, in the vicinity of the resonant frequency, wave nonlinearity is beneficial to the energy conversion performance of the integrated system.

However, this enhancement becomes limited when the wavelength is either excessively large ($kh = 0.44$) or excessively small ($kh = 1.15$). At shorter wave periods, nonlinear effects generate higher-order harmonic wave components with weak penetrability, which are therefore primarily reflected by the outer wall of the OWC chamber, reducing the effective wave–chamber interaction. Conversely, at longer wave periods, a substantial portion of the wave energy tends to bypass the chamber, resulting in reduced energy capture by the OWC system.

Fig. 15 compares the spatial distributions of the free surface elevation for $kh = 0.44$, $kh = 0.50$ and $kh = 1.15$ at the wave height of $H = 0.100$ m. It should be noted that the apparent differences in spatial scale among these plots arise from the fact that the length of the numerical wave flume is adjusted according to the wavelength λ, extending to 4λ upstream and downstream of the structure, while the flume width is kept constant at five times the OWC width. At the resonant frequency ($kh = 0.50$), a pronounced wave focusing effect is observed, characterised by a significant amplification of the free surface elevation within the OWC chamber. The peak wave elevations are located inside the chamber, indicating effective interaction between the incident waves and the internal water column. In contrast, at $kh = 0.44$, although an increase in wave height is observed in the vicinity of the OWC, the maximum free surface elevation occurs predominantly over the annular ramp outside the chamber rather than within it. The excitation of the internal water column is relatively weak, leading to limited energy conversion

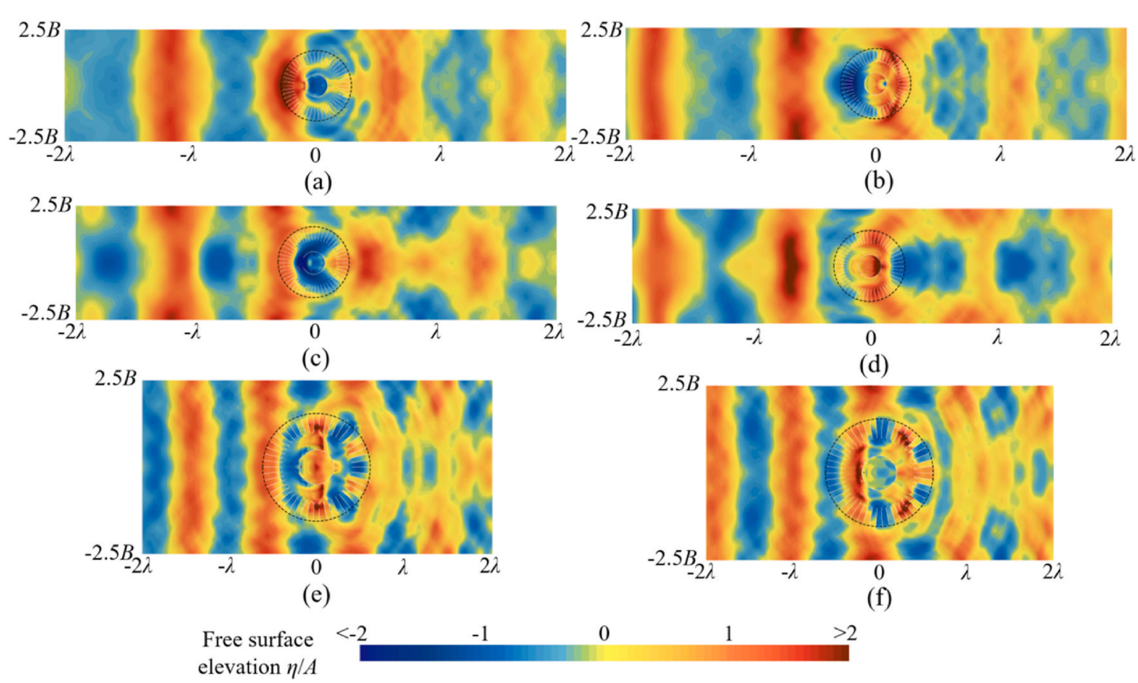

Fig. 15. Comparison of the free surface elevation distributions for $H = 0.100$ m: (a) at $kh = 0.44$, $t = 20.0\ T$; (b) at $kh = 0.44$, $t = 20.5\ T$; (c) at $kh = 0.50$, $t = 20.0\ T$; (d) at $kh = 0.50$, $t = 20.5\ T$; (e) at $kh = 1.15$, $t = 20.0\ T$; (f) at $kh = 1.15$, $t = 20.5\ T$.

efficiency. At $kh = 1.15$, the wave amplification effect becomes minimal. Instead, pronounced wave scattering and reflection occur near the chamber wall, suggesting that the incident wave energy is redistributed into scattered wave components rather than being effectively transmitted into the chamber.

Fig. 16 further illustrates the influence of wave height on the internal hydrodynamic response of the OWC chamber by presenting the variations in the amplitudes of the free surface elevation and the air pressure. It can be clearly seen that, within the range of $kh = 0.44$–0.58, the free surface elevation inside the chamber remains significantly amplified relative to the incident wave height, with an amplification factor generally exceeding 1.5. This indicates that the wave focusing effect persists over the range of wave heights considered. However, this effect gradually diminishes as the wave height increases. In contrast, the air pressure amplitude increases consistently with increasing wave height for all cases. Because the growth in air pressure amplitude outweighs the reduction in wave height amplification, the hydrodynamic efficiency ultimately increases in most cases as the wave height increases. At $kh = 0.44$ and $kh = 1.15$, however, the increase in air pressure amplitude is insufficient to compensate for the pronounced reduction in wave height amplification, leading to a limited efficiency enhancement.

Fig. 17 presents the time histories of the averaged free surface elevation and chamber air pressure at $kh = 0.50$ under two wave heights. With increasing wave height, the free surface elevation exhibits more pronounced deviations from a sinusoidal profile, characterised by sharper crests and shallower troughs, indicating enhanced nonlinear effects. In contrast, under the smaller wave height, the waveform remains relatively smooth and closer to a linear response. The chamber air pressure also shows significant asymmetry. As the wave height increases, the pressure peaks become sharper and more pronounced, whereas the troughs vary more gradually. This behaviour can be attributed to the nonlinear interaction between the oscillating free surface and the airflow through the chamber opening. Despite the increasing nonlinearity with wave height, the phase relationship between the free surface elevation and chamber pressure remains largely consistent, facilitating the increase in efficiency observed in Fig. 14.

3.4. Effects of baffle configuration

Previous studies have suggested that the number of baffles in the wave concentrator may influence wave field transformation (Cen et al., 2024b). Accordingly, seven baffle configurations are considered in the present study to investigate the effects of baffle arrangement on the hydrodynamic efficiency of the WC–OWC integrated system. Specifically, the number of baffles is varied as $N = 0$, 10, 20, 30, 40, 50 and 60. In all cases, the baffles are uniformly distributed with equal angular spacing around the OWC chamber. The corresponding geometric configurations of the integrated system are illustrated in Fig. 18, where the incident wave direction is indicated by the red arrows.

Figs. 19 and 20 present the hydrodynamic efficiency of the WC–OWC integrated system and the corresponding relative efficiency increase for different numbers of baffles under three wave height conditions. As shown in Fig. 19, with relatively few baffles ($N \leq 30$), for all wave heights considered, the hydrodynamic efficiency generally increases with the number of baffles. This indicates that increasing the number of baffles can enhance the wave focusing and flow channelling effects around the OWC chamber, thereby promoting more effective energy capture. However, this efficiency gain gradually levels off when the number of baffles exceeds approximately $N = 30$–40, suggesting the existence of a critical baffle number, beyond which further increases in baffle number yield only marginal additional benefits. For example, when N increases from 30 to 60, the hydrodynamic efficiency increases by only 2.7%–5.8%, while the additional increase from $N = 40$ to 60 is less than 1.5% under the investigated wave height conditions.

A consistent dependence on wave height is also observed. For a given baffle configuration, higher wave heights generally result in higher hydrodynamic efficiency, with the case of $H = 0.100$ m yielding the highest efficiency ($\xi = 0.63$) across the entire range of N. This trend reflects the beneficial role of wave nonlinearity in enhancing the excitation of the internal water column when sufficient wave focusing is achieved. However, as discussed in the previous section, further increases in wave height can induce pronounced wave breaking and associated energy dissipation, which can offset the benefits of nonlinear excitation. Therefore, the observed improvement in hydrodynamic efficiency is valid within the moderate wave height range investigated herein.

Fig. 20 further quantifies the influence of wave height and baffle configuration by illustrating the relative increase in hydrodynamic efficiency with respect to the baseline case of $H = 0.050$ m. It can be seen that the relative efficiency gain initially decreases as the number of baffles increases from $N = 0$ to $N = 20$, followed by a gradual recovery for larger values of N. This suggests that, at low baffle numbers, wave nonlinearity plays a dominant role in enhancing energy conversion, whereas at intermediate baffle densities, the enhancement in wave focusing provided by the guiding baffles offsets the reduced contribution from wave nonlinearity. As the number of baffles further increases, the combined effects of enhanced wave energy concentration and nonlinear wave excitation lead to more pronounced improvements in hydrodynamic efficiency. Overall, these results indicate that both wave height and baffle configuration play important and interdependent roles in determining the energy conversion performance of the WC–OWC integrated system.

To gain further insight into the physical mechanisms responsible for the efficiency trends observed in Figs. 19 and 20, Fig. 21 compares the spatial distributions of the free surface elevation for $N = 0$, 10 and 60. When no baffles are present ($N = 0$), the wave refraction induced by the annular platform is minimal since the motion of fluid particles around the structure remains largely unregulated. As a result, wave energy is not

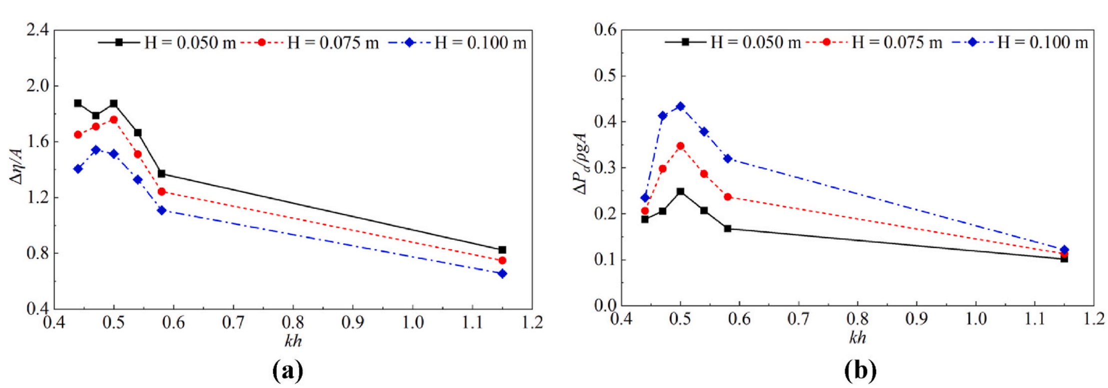

Fig. 16. Variations of the free surface elevation amplitude (a) and the air pressure amplitude (b) inside the chamber for $H = 0.050$ m, $H = 0.075$ m and $H = 0.100$ m.

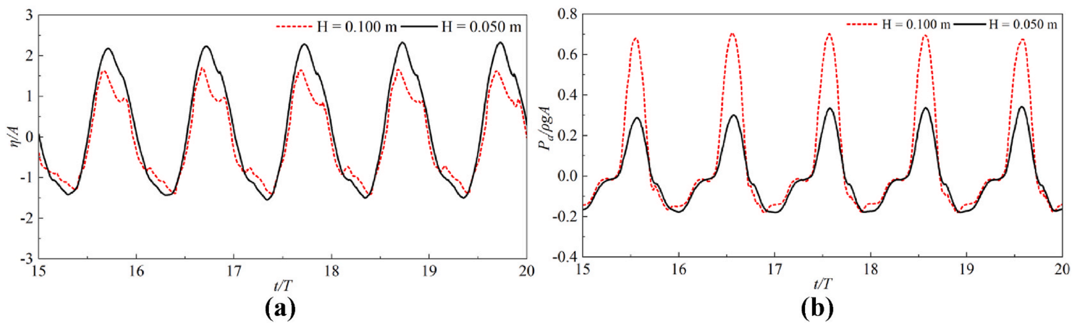

Fig. 17. Time histories of the averaged free surface elevation (a) and air pressure (b) inside the chamber at $kh = 0.50$ for $H = 0.050$ m and $H = 0.100$ m.

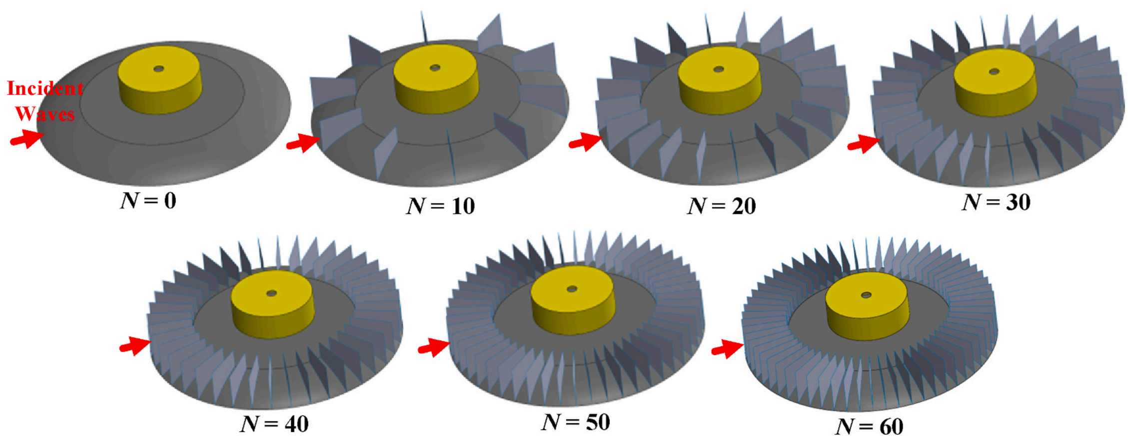

Fig. 18. Geometric configurations of the WC–OWC integrated system with different numbers of baffles.

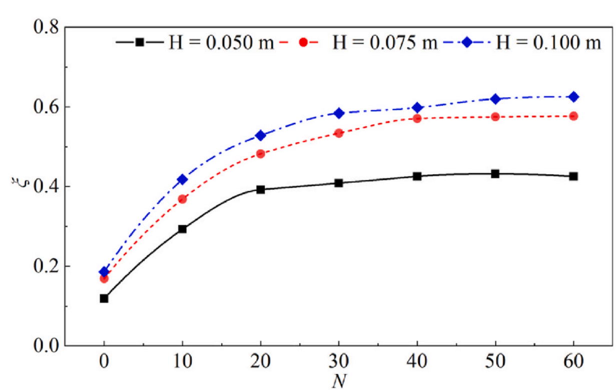

Fig. 19. Hydrodynamic efficiency of the WC–OWC integrated system for different number of baffles.

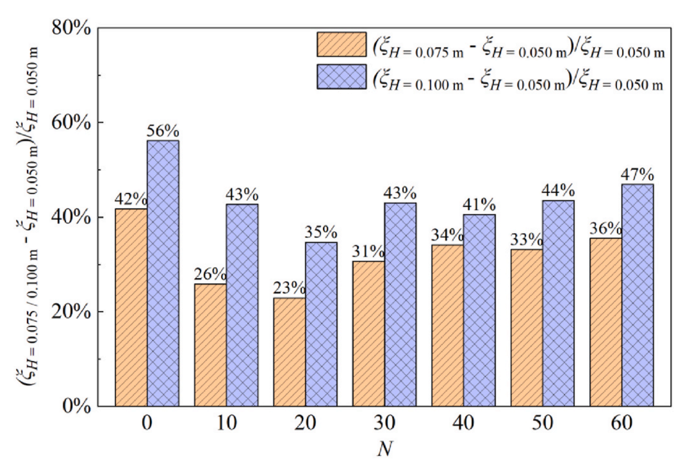

Fig. 20. Relative increase in hydrodynamic efficiency of the WC–OWC integrated system for different number of baffles.

effectively guided toward the OWC chamber and instead tends to propagate past the structure, with only a slight increase in wave height observed in the downstream region rather than within the chamber. As the number of baffles increases, the baffles progressively impose an effective channelling effect on the local flow field. Fluid particles are guided through the gaps between adjacent baffles in the radial direction of the circular platform, which enhances wave refraction and induces increasing curvature of the wavefronts. Consequently, wave energy is gradually redirected toward the centre of the structure and effectively transmitted into the chamber, which explains the improved efficiency at higher baffle densities.

Fig. 22 depicts the internal hydrodynamic response of the OWC chamber. The free surface elevation amplitude generally increases with the number of baffles but gradually approaches a plateau, or exhibits a slight decrease, as N becomes large, indicating that the wave height amplification reaches a limiting level. For all baffle configurations, the wave height amplification inside the chamber decreases with increasing incident wave height, with the maximum amplification factor reducing from approximately 1.9 for $H = 0.050$ m to about 1.5 for $H = 0.100$ m. In contrast, the air pressure amplitude increases consistently with both increasing wave height and baffle number, which ultimately supports the overall enhancement in hydrodynamic efficiency observed in Fig. 19.

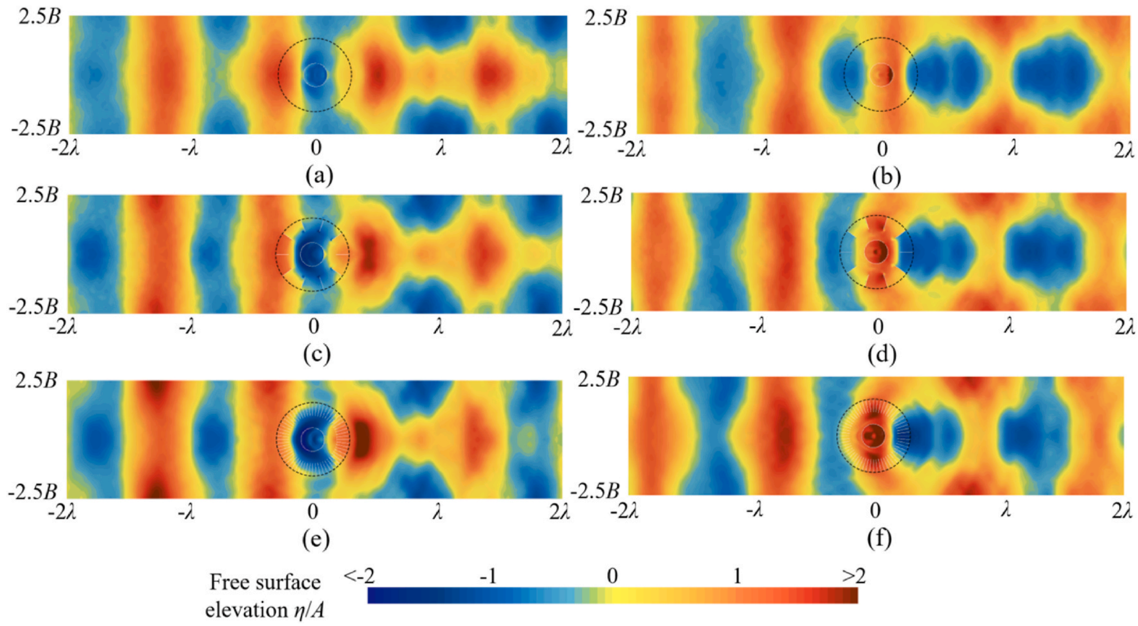

Fig. 21. Comparison of the free surface elevation distributions for $H = 0.050$ m: (a) at $N = 0$, $t = 20.0$ T; (b) at $N = 0$, $t = 20.5$ T; (c) at $N = 10$, $t = 20.0$ T; (d) at $N = 10$, $t = 20.5$ T; (e) at $N = 60$, $t = 20.0$ T; (f) at $N = 60$, $t = 20.5$ T.

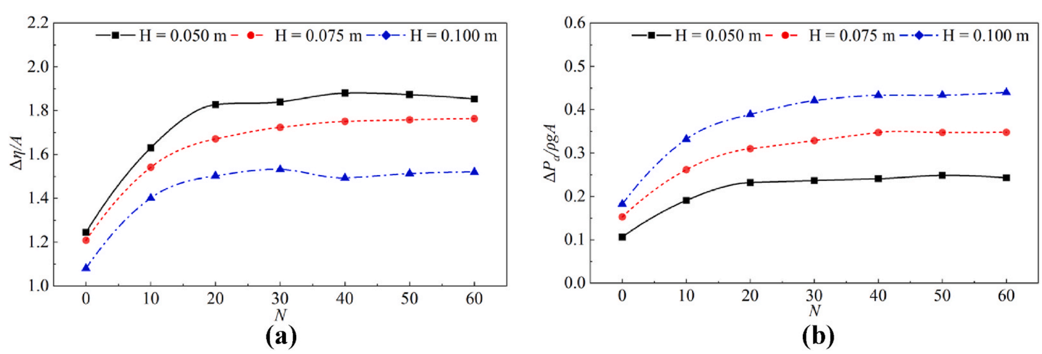

Fig. 22. Variations of the free surface elevation amplitude (a) and the air pressure amplitude (b) inside the chamber for different number of baffles.

4. Conclusions

This study investigates the wave energy capture performance of a novel axisymmetric OWC WEC integrated with an annular wave concentrator. A three-dimensional CFD model is developed and validated against experimental data, demonstrating its capability of accurately resolving the coupled hydrodynamic and pneumatic responses of the integrated system. A systematic comparison between the standalone OWC device and the proposed WC–OWC configuration is carried out over a broad range of regular wave periods, followed by parametric investigations into the effects of wave height and baffle configuration on the hydrodynamic efficiency.

By integrating the annular wave concentrator, the proposed WC–OWC system effectively captures the energy of long-period waves that would otherwise bypass the standalone OWC, resulting in a 77.3% increase in peak hydrodynamic efficiency. This enhancement can be attributed to the wave refraction and superposition within the concentrator, which amplify the free surface variation inside the chamber to approximately twice the incident wave amplitude and thus significantly increase the chamber air pressure. The associated low-frequency shift of the resonant response, with the resonance peak changing from $kh = 1.39$ to $kh = 0.50$, leads to the relative efficiency increase by approximately 300%–1000% with the wave period ranging from 2.8s to 3.8 s, thereby altering the effective operating range towards longer-period waves.

Wave nonlinearity is found to play a beneficial role in energy conversion near the resonant condition, yielding an additional 43.5% increase in CWR as the wave height increases from 0.050 m to 0.100 m under non-breaking wave scenarios. Although the amplification of the free surface elevation oscillation weakens with the increasing wave height, the amplitude of the pressure fluctuation keeps rising, leading to an overall enhancement of hydrodynamic efficiency. However, this effect becomes limited under excessively long or short wave conditions, where the incident wave either bypasses the chamber or is converted into higher-order harmonic components with weak penetrability, which are consequently reflected by the chamber wall.

With relatively few baffles, increasing the number of the guiding baffles significantly enhances the hydrodynamic efficiency by strengthening wave focusing and flow channelling towards the OWC chamber, while the efficiency gain gradually levels off when a critical baffle number of 30–40 is reached. In particular, in the absence of baffles, minimal wave refraction and unregulated flow allow a substantial portion of the incident wave energy to bypass the chamber, leading to inefficient energy capture.

Overall, this study demonstrates that the proposed axisymmetric WC–OWC configuration provides an effective and wave-direction-independent approach to enhancing wave energy conversion. The physical insights gained into the effects of wave focusing, nonlinearity, and guiding baffles offer valuable guidance for the design and optimisation of the next-generation OWC-based wave energy converters. Future work will extend the present analysis to irregular wave conditions and incorporate detailed turbine–PTO modelling to assess the overall wave-to-wire performance of the integrated system.

CRediT authorship contribution statement

Yuhao Cen: Writing – original draft, Formal analysis, Conceptualization. **Qianze Zhuang:** Validation, Formal analysis. **Dongfang Liang:** Writing – review & editing, Funding acquisition. **Tahsin Tezdogan:** Writing – review & editing, Methodology. **Xiaodong Liu:** Software, Methodology. **Yifeng Yang:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Dongfang Liang reports financial support was provided by The Royal Society. Dongfang Liang reports financial support was provided by National Key Research and Development Program of China. Given his role as the Editor-in-Chief, Professor Tahsin Tezdogan had no involvement in the peer review of this article and had no access to information regarding its peer review. Full responsibility for the editorial process for this article was delegated to another journal editor. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that support the findings of this study are available within the article.

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